



ELEOS Recumbent Off-Road Handcycle

Michael Chase, Christian Kowalski, Sam Manina, Zachary McNees, Dane Ziegler
Advisors: Robert Melendy, Rebekah Slyter

Introduction

ELEOS Solutions is a non-profit organization that seeks to provide engineering solutions to problems faced by underserved communities. People with lower-limb disabilities throughout the world often lack mobility options, but handcycles provide this community with an alternative to bicycles, giving them independence and freedom of movement. ELEOS has sponsored a senior design team to help make effective handcycles more accessible to people who need them.



Left to Right:
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Problem Statement

Current off-road handcycles with integrated power assist are prohibitively expensive, leaving many people with limited freedom of movement. Existing assistive devices often fail on rugged terrain such as dirt, mud, or rocky paths, limiting mobility. To address this gap, this project will design and prototype a recumbent off-road handcycle platform that is durable, affordable, and capable while incorporating power assist with hybrid propulsion.

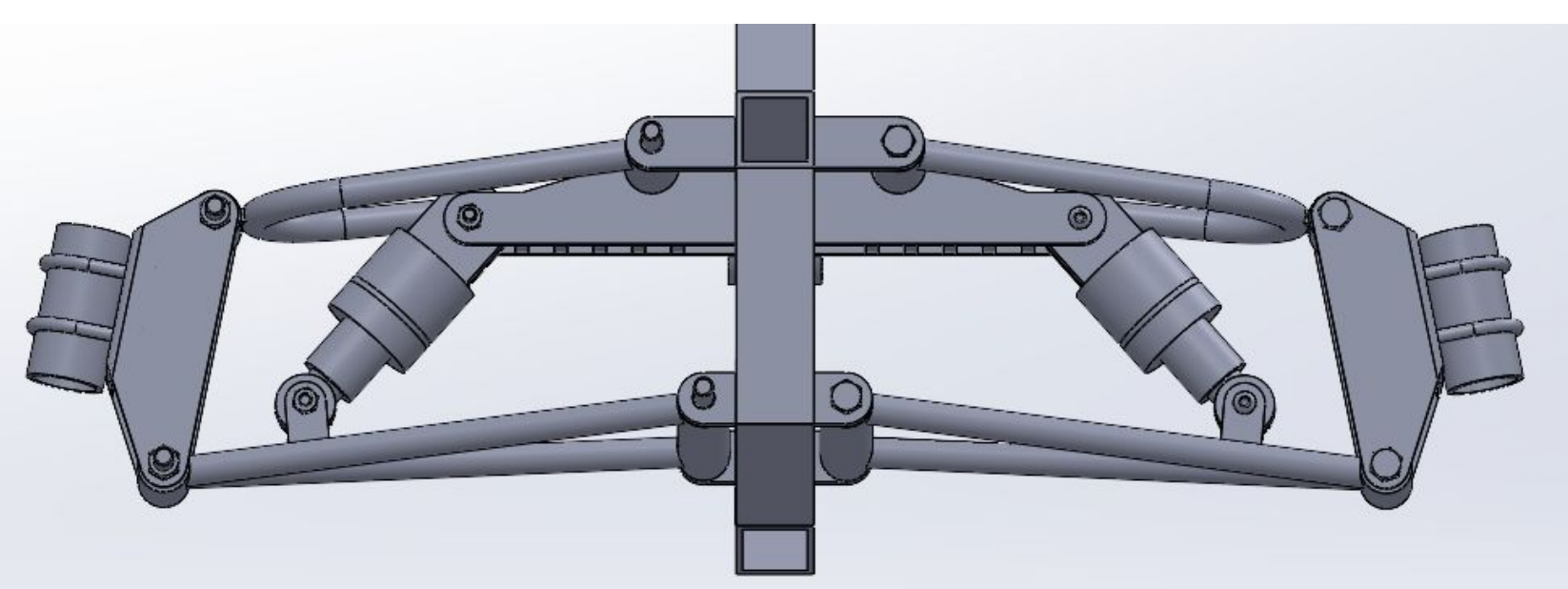
Design Details

1 Steering



While it is possible to steer using the hand cranks, a set of handlebars attached to the steering column provides additional leverage for hard turns as well as a place to mount the brake levers, gear shifter, and e-assist controls. The steering geometry was designed for low effort direct steering with auto-centering front wheels and minimal disturbance from the movement of the suspension.

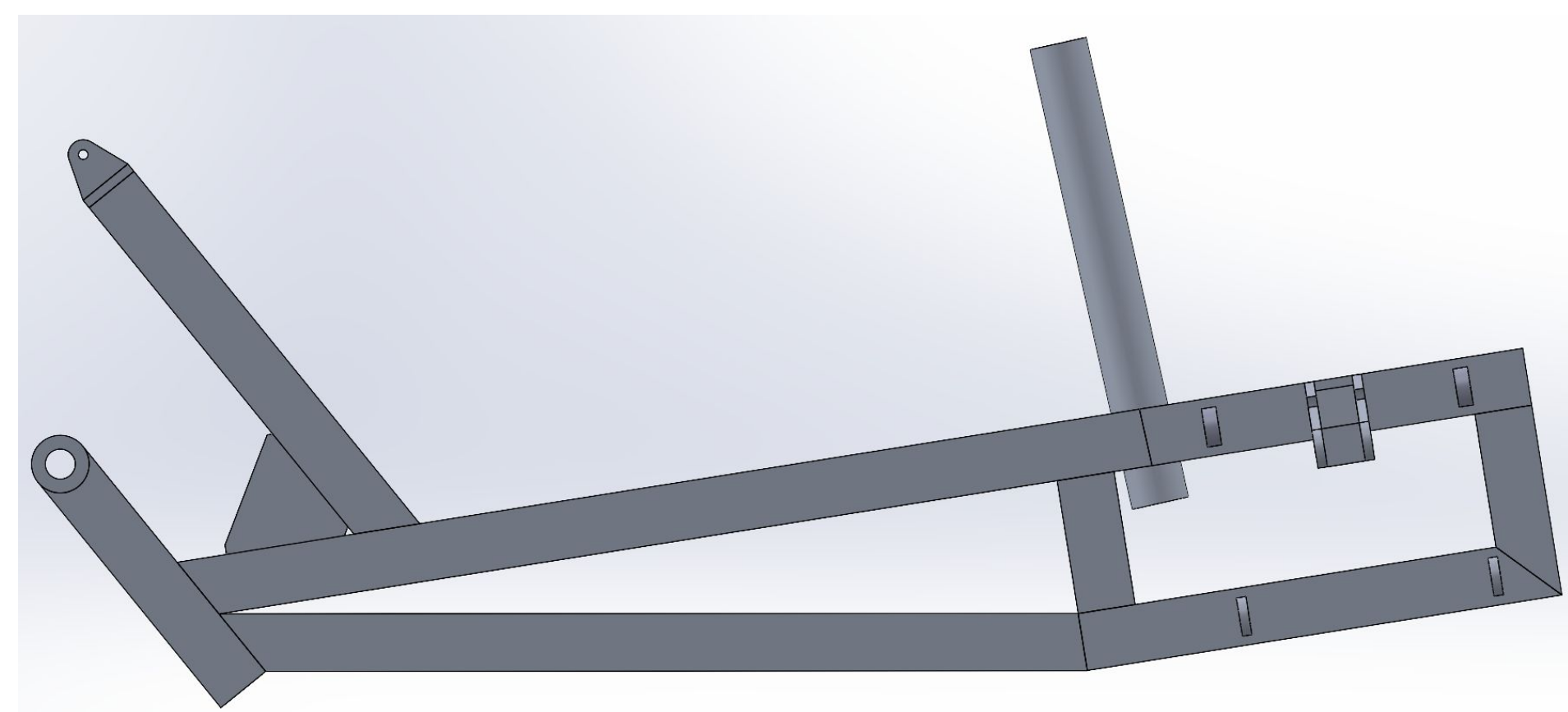
2 Suspension



This handcycle has both front and rear suspension. The front wheels use a "double wishbone" suspension system, with the rear being a traditional bike style suspension. In tandem, these are well suited to absorb large bumps encountered when off-roading.

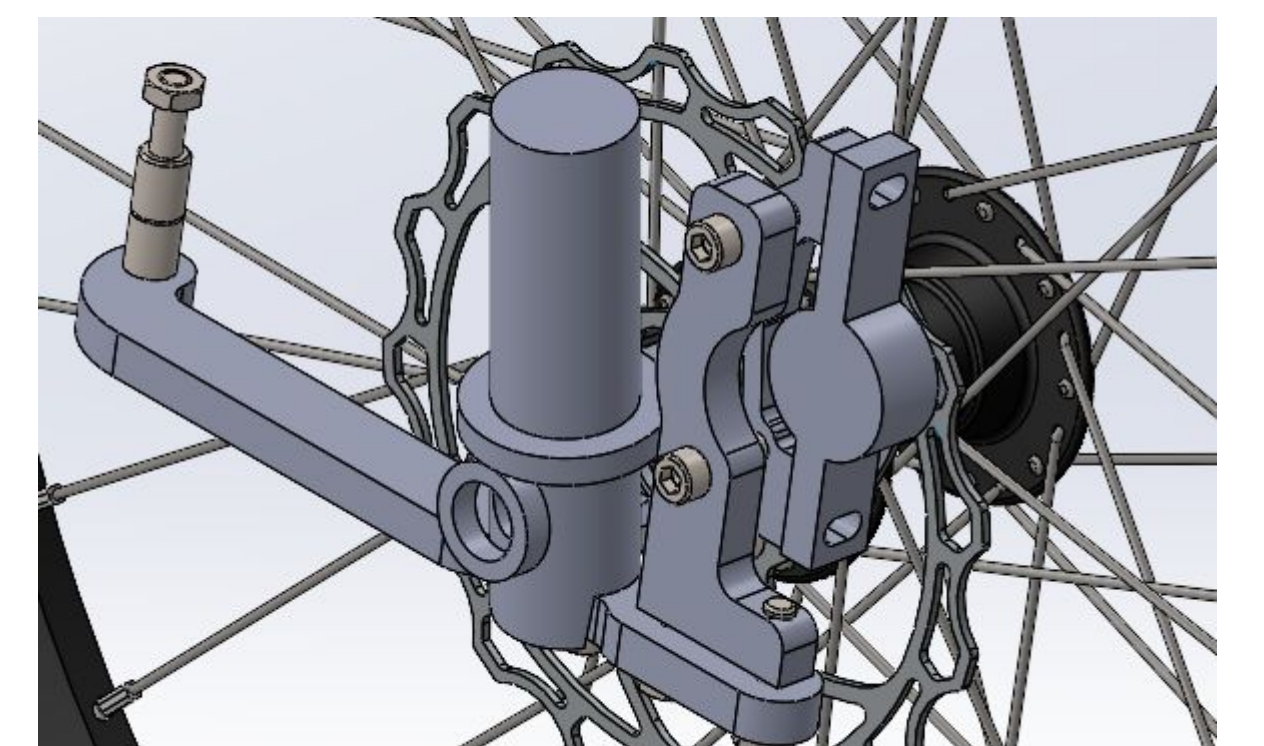


3 Frame



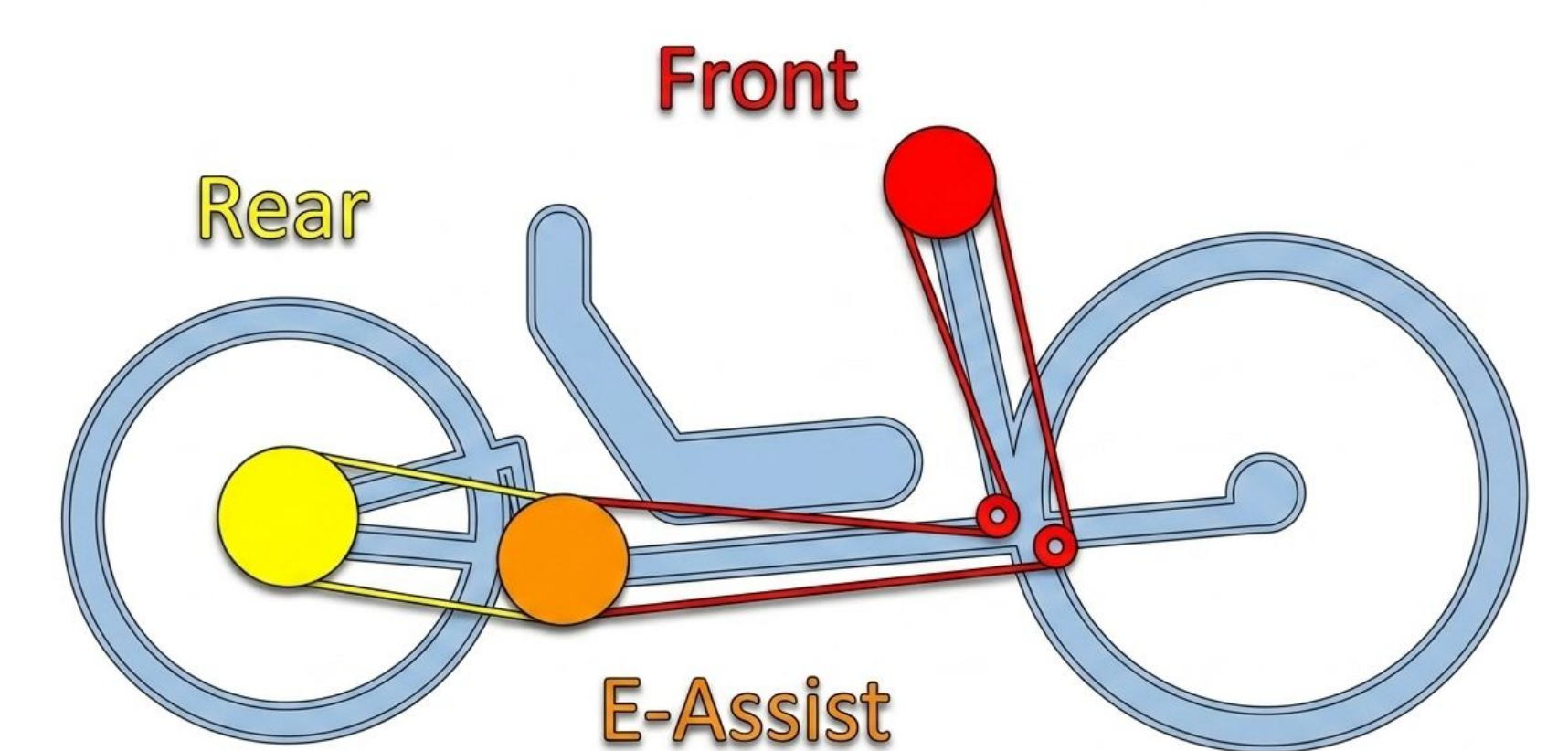
The custom frame is primarily constructed from welded T-6061 aluminium alloy square tube stock. This durable alloy is commonly used in bikes, providing a strong but lightweight base for our handcycle. Mounted directly on the frame is the seat, leg slings, chain tensioner, and routing sprockets.

5 Wheel Mounts



The front wheels are special tricycle wheels that are able to be supported from only one side. This also allows for easy tire replacement during maintenance. The axle is set through the kingpin, which connects the wheels to the suspension system and allows the wheels to pivot as the handcycle is steered. The tie rods connect to a lever arm attached to the kingpin which also supports the disc brake mounts. The rear wheel is mounted on a quick release for easy removal of the rear wheel.

4 Drive Train



The drivetrain has three distinct sections: the front, the e-assist, and the rear. The front is designed to transfer torque from the hand cranks back to the e-assist while enabling steering during cranking. The e-assist provides extra motor torque to the rear section, which takes the enhanced power from the cranks and drives the rear wheel, accelerating the handcycle. The rear section has a 1x10 mountain bike gear system to further modify the input power of the rider to suit the terrain being traversed. The result is a vehicle that can handle rough ground and hills using only hand cranking or throttle input.

Conclusion

Our fully-functional prototype was built using a wide variety of manufacturing techniques including welding, milling, and waterjet cutting and came to a final material cost of ~\$3,050. While this price point is still inaccessible to many, it is nonetheless significantly cheaper than existing solutions on the market. Further optimizations may be able to reduce the price while improving performance. A team next year will take our design and iterate on it, testing its limits and optimizing it in order to fully achieve the goal of a durable, affordable, and capable offroad handcycle.

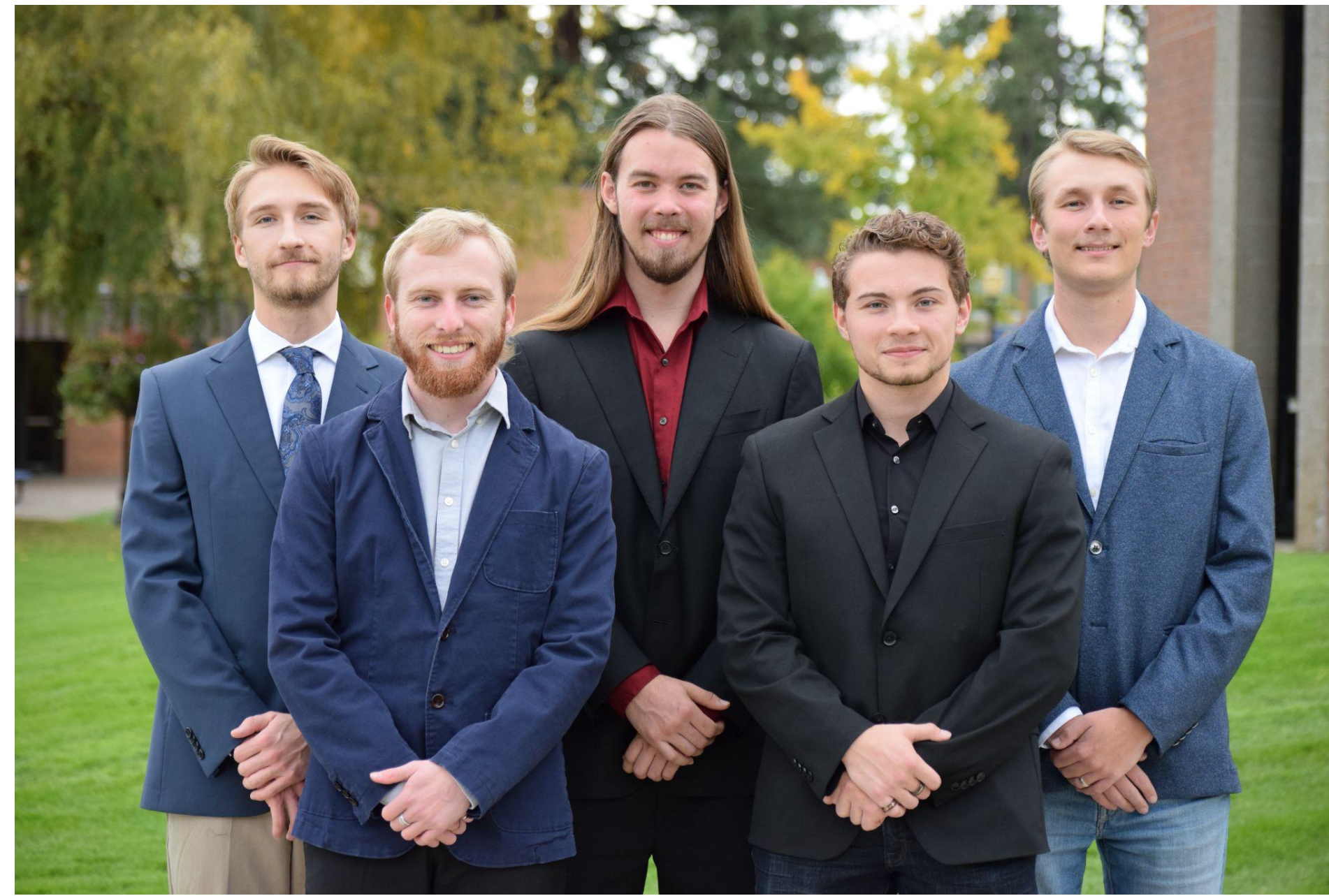


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Problem Statement

Current off-road handcycles with integrated power assist are prohibitively expensive, leaving many people without access to durable mobility solutions. Existing assistive devices often fail on rugged terrain such as dirt, mud, or rocky paths, limiting opportunity. To address this gap, this project will design and prototype a recumbent off-road handcycle platform that is both affordable and adaptable, incorporating power assist with hybrid propulsion.

Design Details

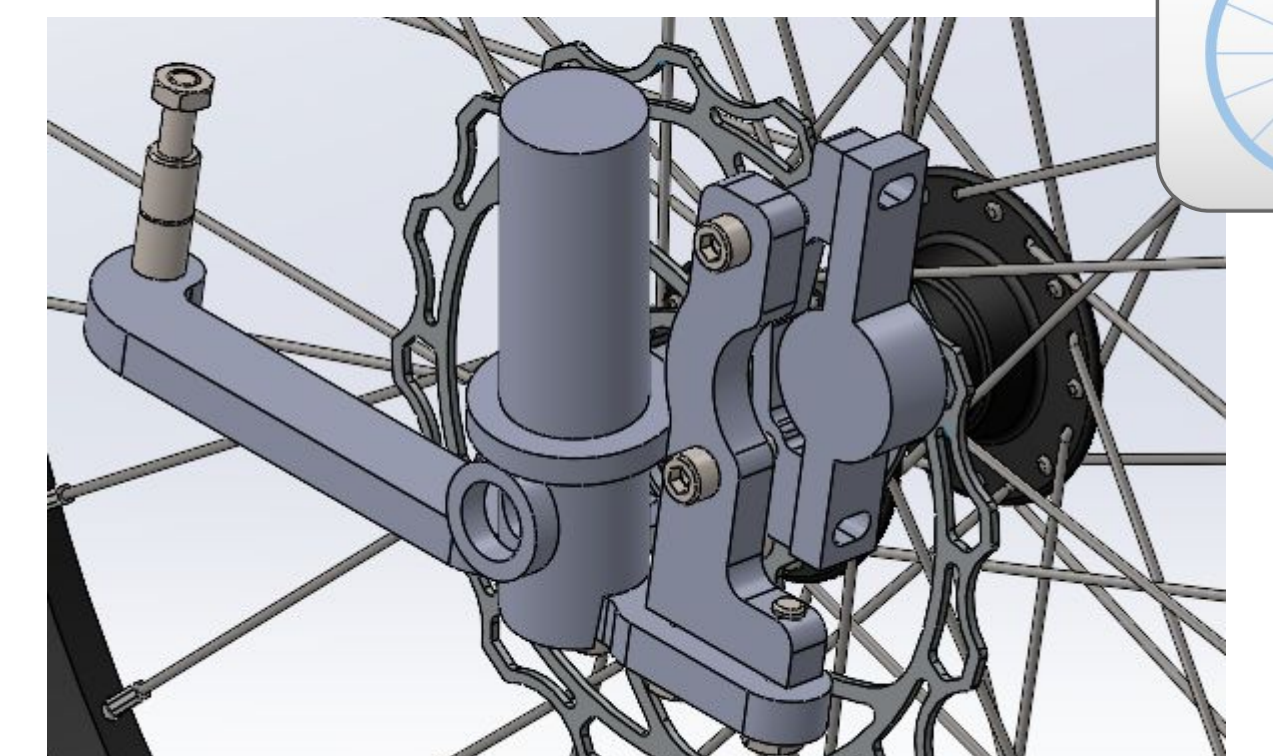
Steering



While it is possible to steer using the hand cranks, a set of handlebars mounted on the steering column provides additional leverage for hard turns as well as a place to mount the brake levers, gear shifter, and e-assist screen. The steering geometry was designed for low effort direct steering with auto-centering front wheels and minimal disturbance from the movement of the suspension.

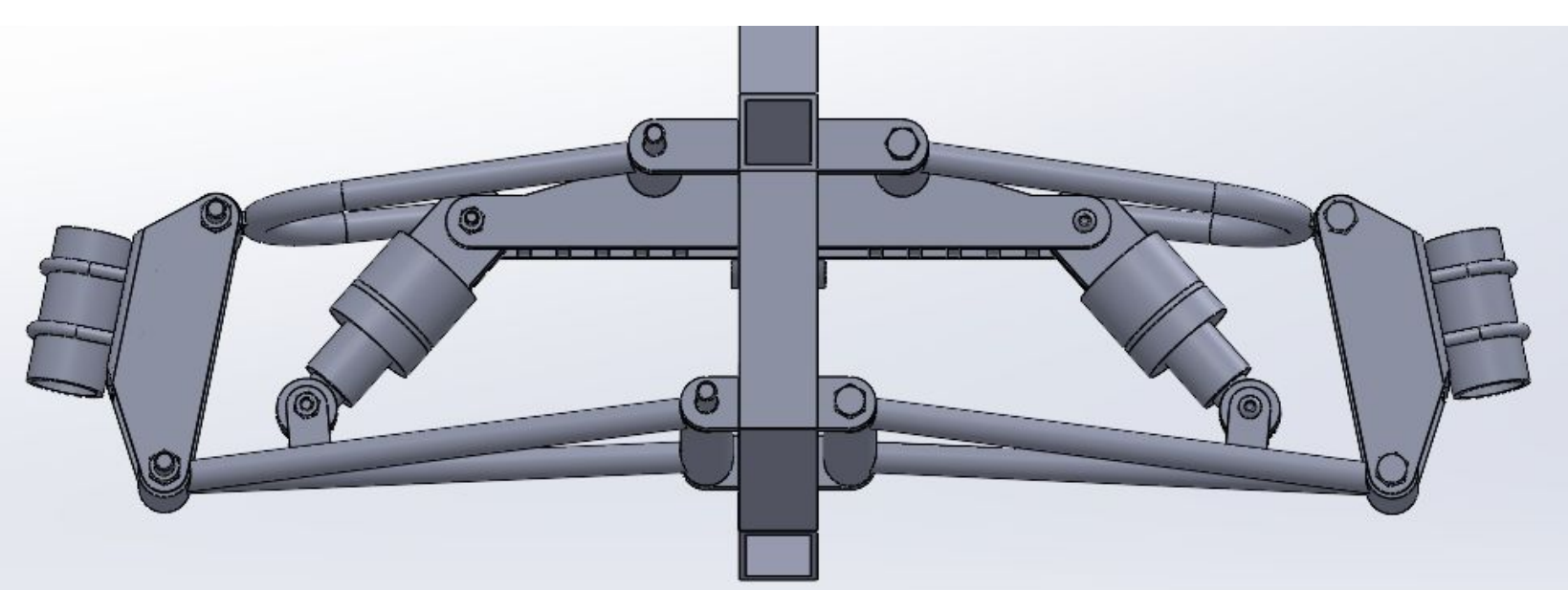


Wheel Mounts



The front wheels are special tricycle wheels that are able to be supported from only one side. The axle is set through the kingpin, a cylinder that connects the wheels to the suspension system and allows the wheels to pivot as the handcycle is steered. The tie rods connect to a lever arm attached to the kingpin which also supports the disc brake mounts. This allows for easy tire replacement during maintenance. The rear wheel mount is a traditional bike quick release, for easy removal of the rear wheel.

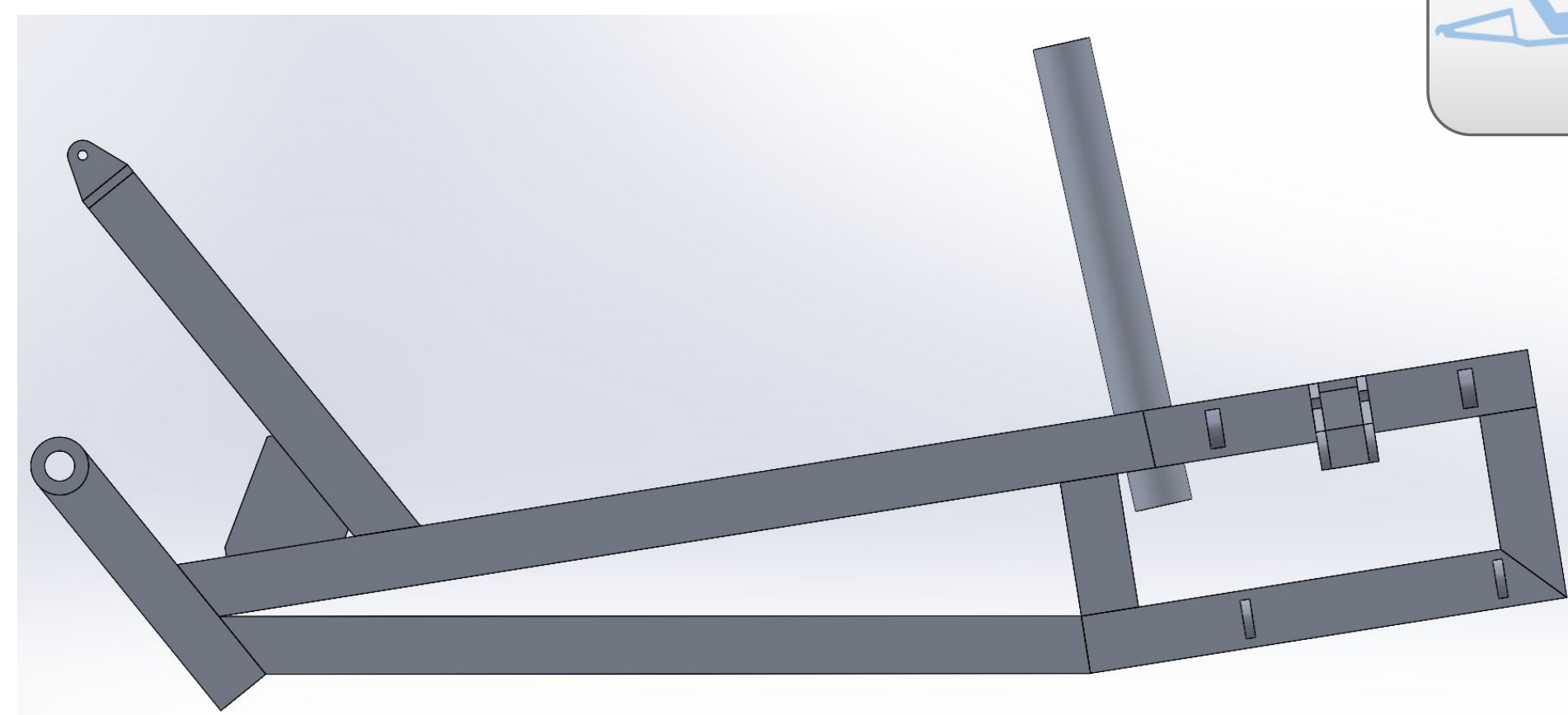
Suspension



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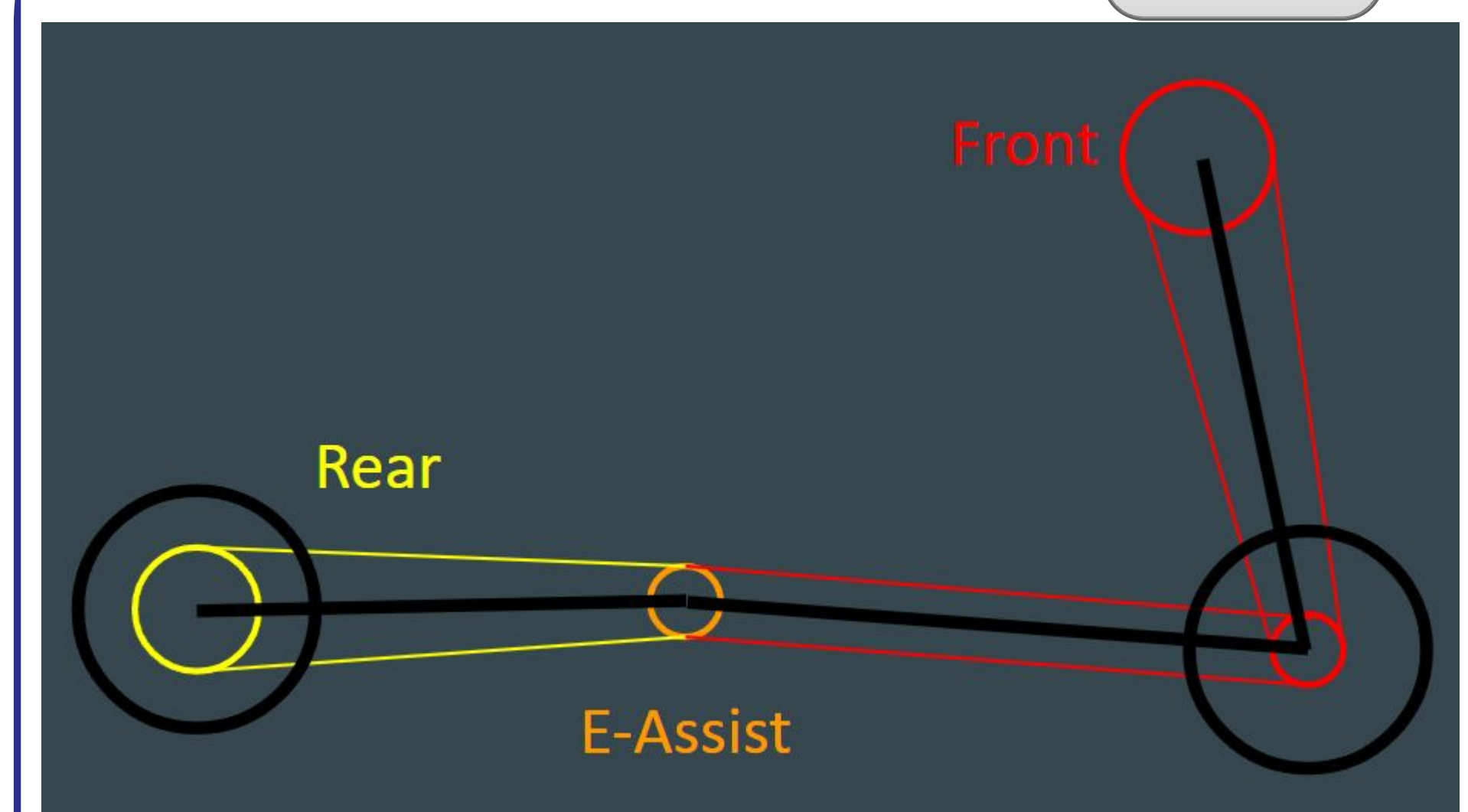


Frame



The frame is primarily constructed from welded T-6061 aluminium square tube stock. This durable alloy is commonly used in bikes, providing a strong but lightweight base for our handcycle. The seat, leg slings, and routing sprockets.

Drive Train



The drivetrain has three distinct sections: the front, the e-assist, and the rear. The front transfers torque from the hand cranks back to the e-assist. The e-assist provides extra motor torque to the user's cranking and transfers the total power to the rear section. The rear takes the enhanced power from the cranks and drives the rear wheel, accelerating the handcycle. The rear section has a 1x10 mountain bike gear system to further modify the input power of the rider to suit the terrain being traversed. This drivetrain system allows the user to pedal with out the use of their legs, and get help from the e-assist when pedaling up steep terrain.

Conclusion

A senior design team next year will take our design and iterate on it, testing its limits and optimizing it in order to fully achieve the goal of a durable, affordable, and capable offroad handcycle. Our fully-functional prototype was built using a wide variety of manufacturing techniques including welding, milling, and waterjet cutting and came to a final material cost of ~\$3,500. While this price point is still inaccessible to many, it is nonetheless significantly cheaper than existing solutions on the market. Further optimizations may be able to reduce the price while improving performance.

Senior Design



Eleos Recumbent Off-Road Handcycle

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Introduction

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Problem Statement

Current off-road handcycles with integrated power assist are prohibitively expensive, leaving many individuals-especially in low- and middle-income regions-without access to durable mobility solutions. Existing assistive devices often fail on rugged terrain such as dirt, mud, or rocky paths, limiting opportunity. To address this gap, this project will design and prototype a recumbent off-road handcycle platform that is both affordable and adaptable, incorporating power assist with hybrid propulsion.

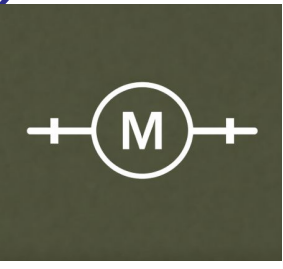
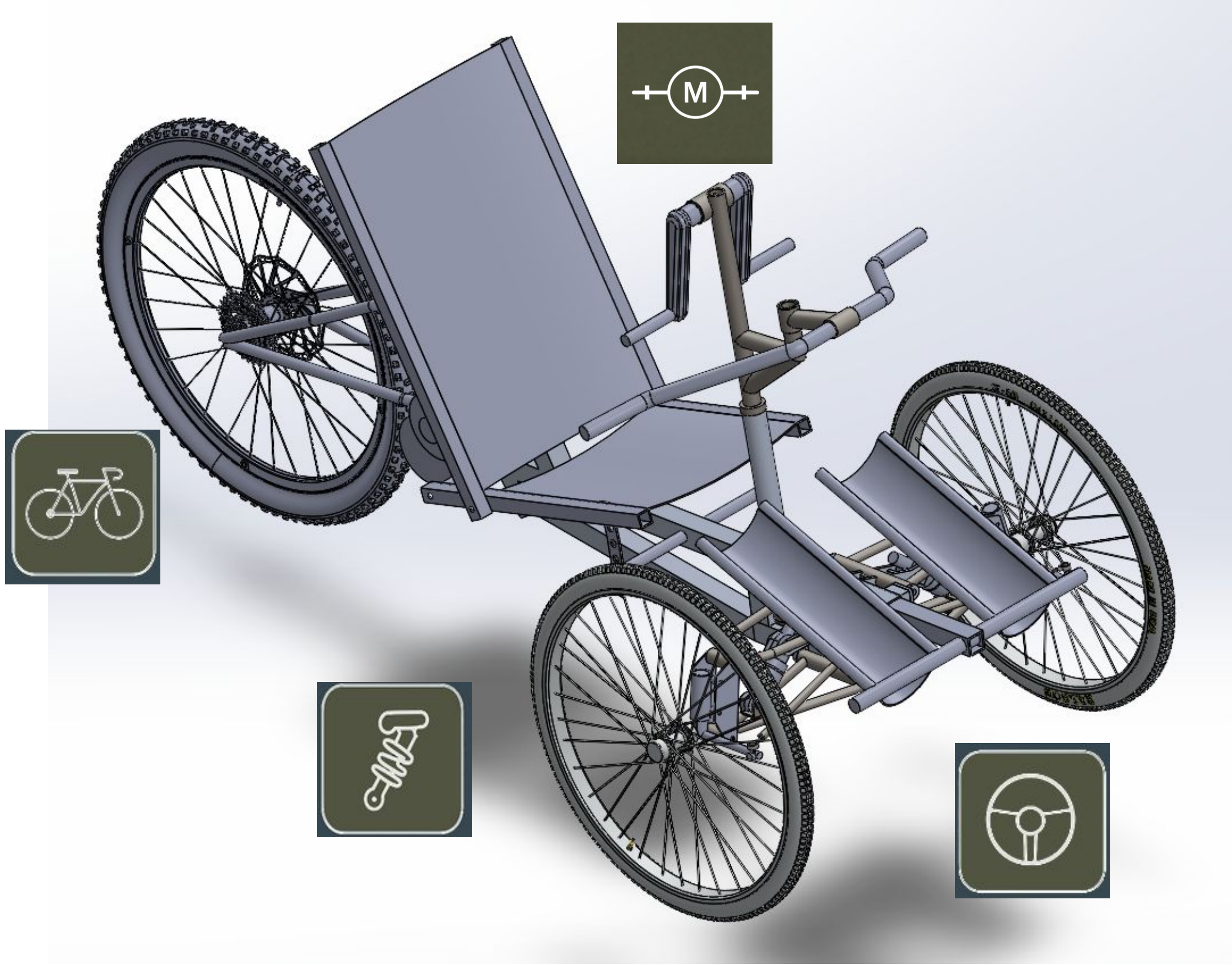
Design Details



DriveTrain



1x10 mountain bike drive system avoids mounting issues of front derailleur while still giving gear options links to front via eassist



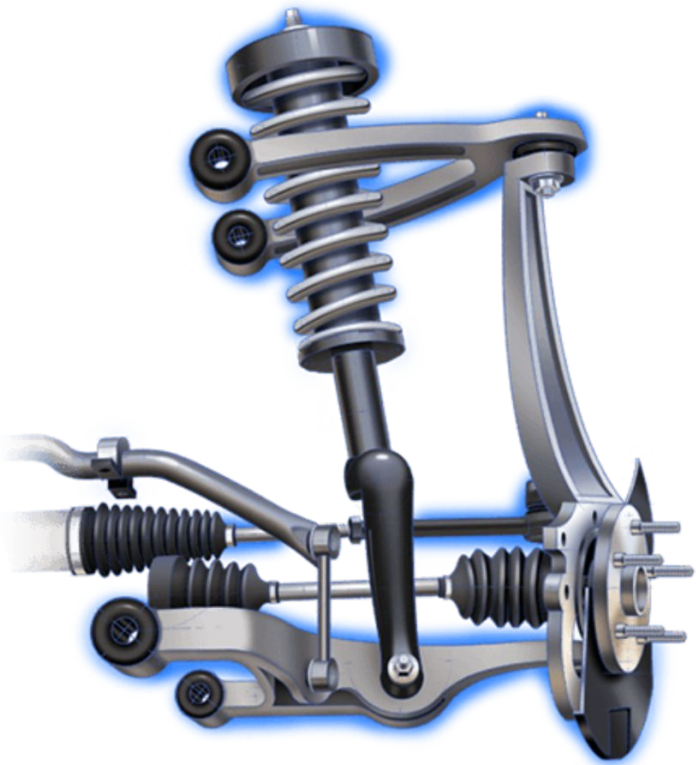
E- Assist



The E-Assist functionality allows the rider to engage a motor to assist them in pedaling. A 750 W motor connected to a ___ aH battery ensures the rider ample assist for long rides. The E-Assist also incorporates an easy to use screen to show battery life, speed, and offers pedal assist modes based on rider preference.

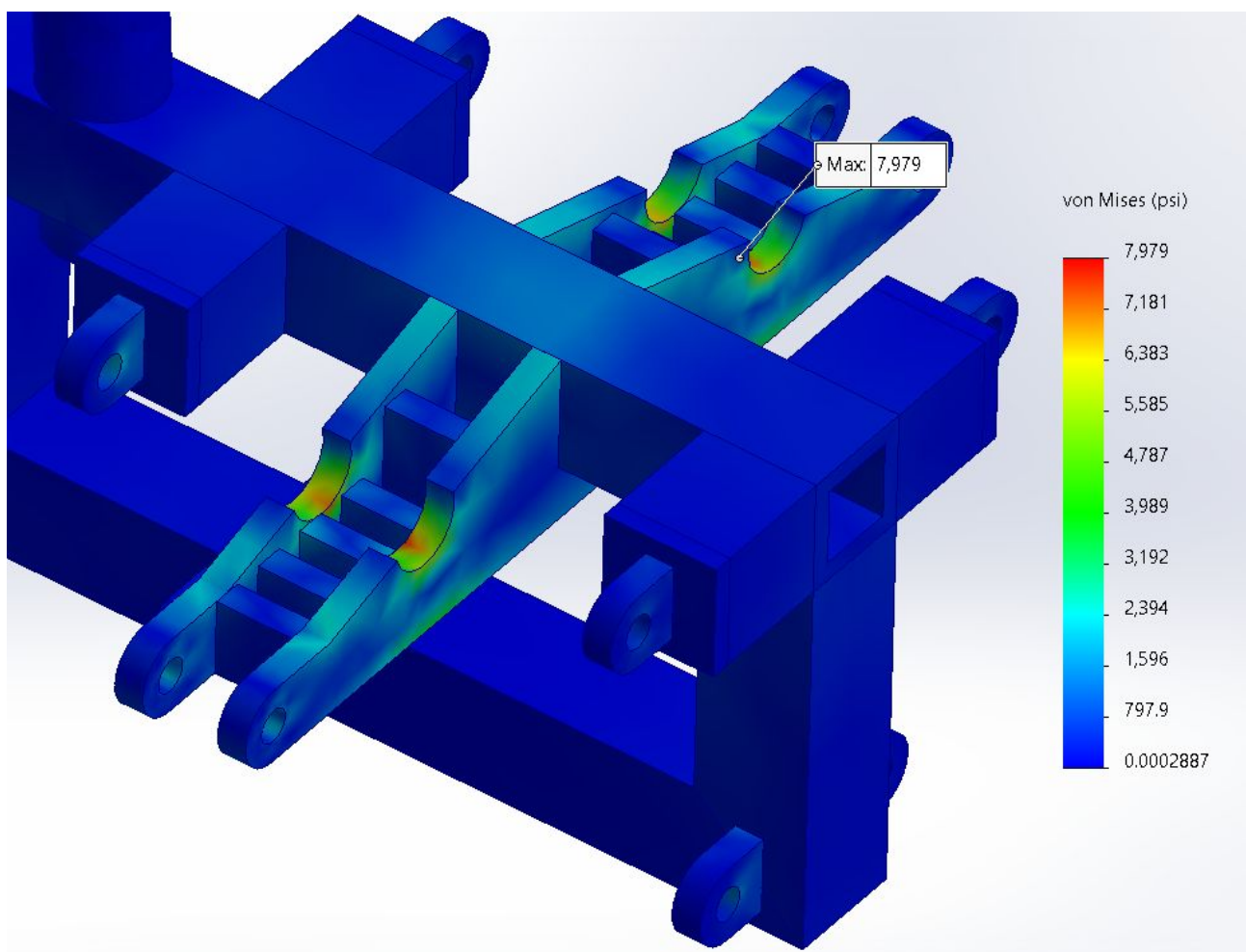


Suspension



We chose to design using a double wishbone suspension system, which consists of two arms that connect to the wheel to the frame, which is then supported by a shock in the center. To keep final manufacturing prices down, we found and designed around some off the shelf stock arms that we purchased for our design. Once we had the geometry, the next step was selecting the right shock...

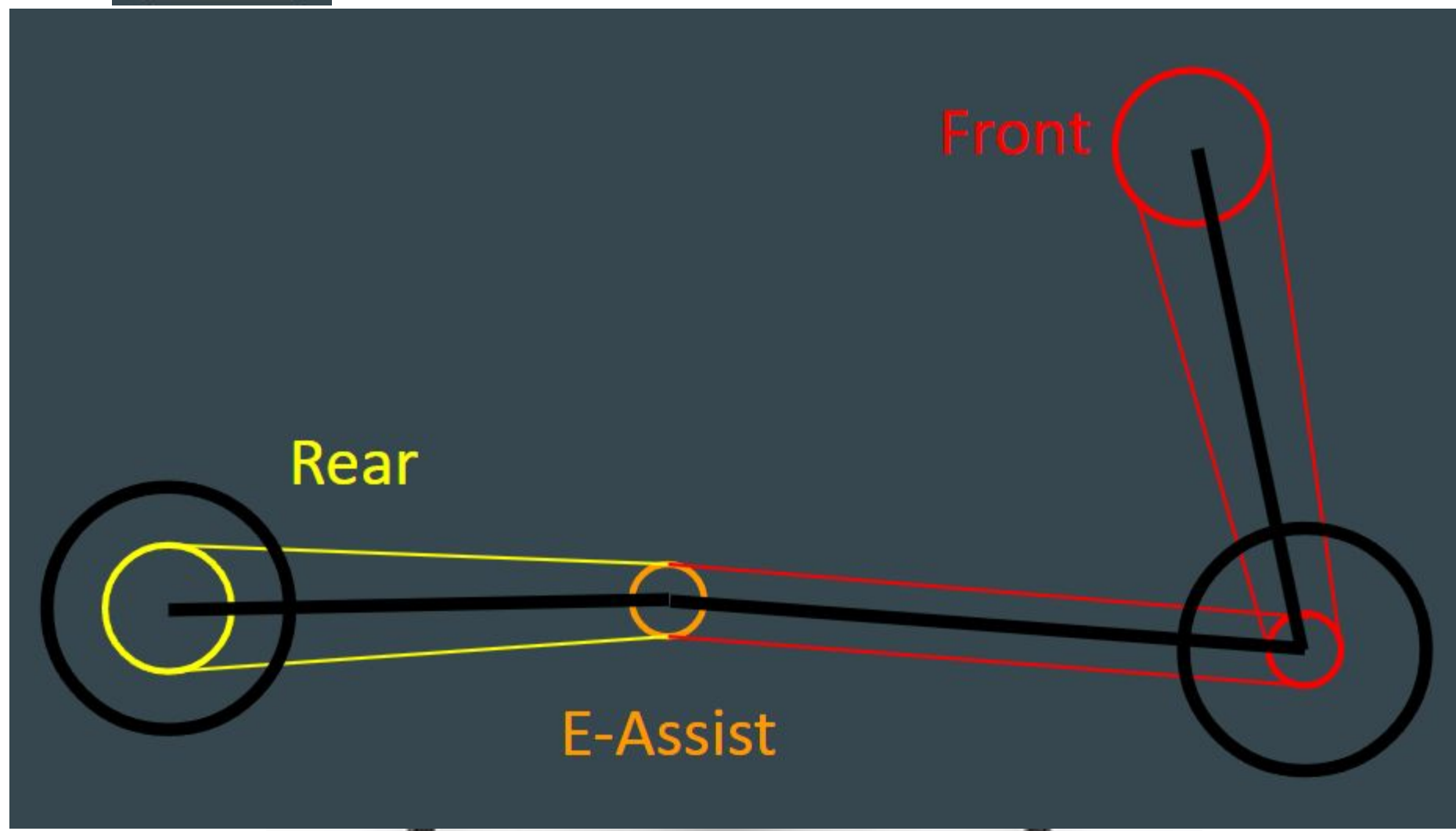
Simulations



To validate the design before fabrication, we ran FEA on this critical component using 6061 aluminum properties. We compared our stress levels to the material yield strength, and verified that we're operating safely below failure limits. With that validation, we moved into manufacturing.



Steering



Placeholder

A handcycle must also be able to be steered while using your arms to turn the cranks. The hand cranks, however, are close together, making it more difficult to steer; therefore, additional handlebars must be added that can be used to steer the bike easier in tight turns and rough terrain. This second set of handlebars will also be where the brakes are. Videos of off-road handcycles in action often show the rider using one hand to crank while the other hand uses the handlebars to steer and brake.

Conclusion

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